## 1. Introduction and Dataset Overview

Under GDPR Article 12, corporate privacy notices *must be communicated in a concise, transparent, intelligible and easily accessible form, using clear and plain language*. This report investigates that requirement empirically by treating it as a measurable question: do privacy notices display the quantifiable surface features of plain-language writing?

To answer this, two small corpora are compared: 10 corporate privacy notices and 10 gov.uk public-information pages on related subject matter (data protection, freedom of information, personal-information charters). The gov.uk pages, produced under the enforced GOV.UK plain-language style guide, serve as a documented reference point for plain-English writing. The comparison rests on the assumption that if privacy notices genuinely met the plain-language standard, their measurable linguistic features should resemble those of the gov.uk pages.

The corpus and feature extraction were developed as part of a separate computational linguistics project on GDPR compliance in privacy notices. The documents were collected and cleaned, and the features extracted computationally. This report uses the resulting dataset (corpus_features.csv) as the starting point and focuses exclusively on its statistical analysis.

This report isolates two numeric features as indicators of complexity and accessibility:

- avg_sent_len (Average Sentence Length): the mean number of words per sentence, serving as a proxy for syntactic complexity.
- flesch_re (Flesch Reading Ease): a standardized readability index where higher scores denote text that is easier to comprehend, and lower scores indicate dense, structurally complex language.

Using descriptive statistics, frequency distributions, and confidence intervals, this report quantifies and compares the readability and sentence structure of the two groups.

## 2. Descriptive Statistics

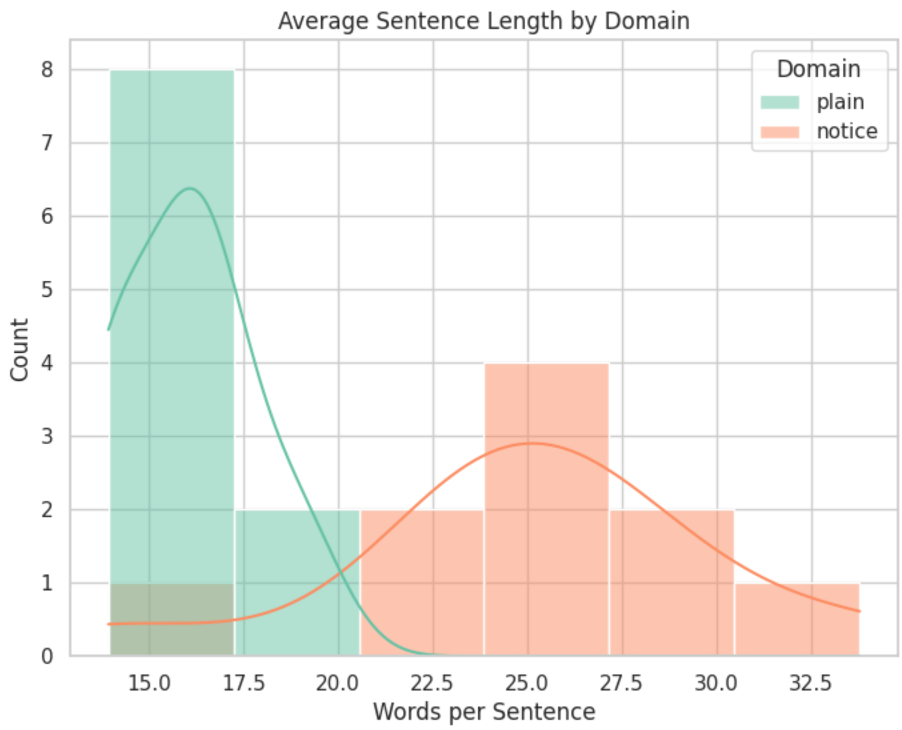
To summarise the central tendency and variability within each domain, the mean, median, standard deviation (SD), and interquartile range (IQR) were computed for both features. The distributions were later visualised using histograms and boxplots.

| **Domain** | **avg_sent_len** | | | | **flesch_re** | | | |
|---|---|---|---|---|---|---|---|---|
| | Mean | Median | SD | IQR | Mean | Median | SD | IQR |
| **Notice** | 25.01 | 24.56 | 4.95 | 3.64 | 34.52 | 32.59 | 7.30 | 8.22 |
| **Plain** | 16.03 | 16.14 | 1.76 | 2.25 | 50.61 | 49.71 | 7.38 | 12.80 |

### 2.1. Average Sentence Length

In case of syntactic complexity, clear structural variations exist between the two groups. Documents in the “notice” domain exhibit substantially longer sentences on average (Mean = 25.01 words) compared to the “plain” domain (Mean = 16.03 words). Furthermore, the “notice” texts show a much higher standard deviation (4.95), indicating that their sentence lengths vary more widely than “plain” texts (SD = 1.76). However, the interquartile range shows a smaller gap (3.64 vs 2.25). This suggests that the bulk of “notice” texts are fairly consistent in sentence length, and that the larger SD might be driven by a few documents with unusually high or low values.

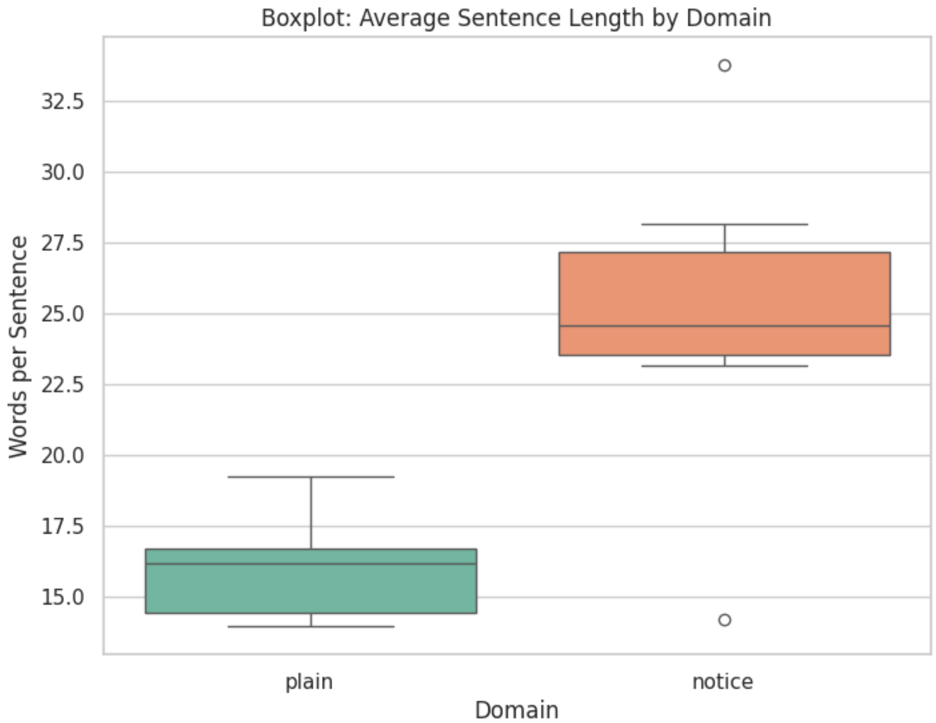
The accompanying histogram visually reinforces these findings. The “plain” language dataset forms a steep, tightly packed peak concentrated between 14 and 17 words per sentence. This narrow grouping visually confirms its low standard deviation and reflects a consistent adherence to concise phrasing. In contrast, the “notice” domain displays a much broader and flatter distribution shifted to the right. While its most frequent sentence length falls around the 24 to 27-word mark, the data extends up to roughly 34 words - a higher variability that matches its larger standard deviation. The clear separation between the two peaks indicates that official notices rely on longer, more complex sentence structures, whereas “plain” texts maintain shorter, more uniform sentences.




The boxplot further confirms the earlier observations. In a boxplot, the coloured box shows where the middle 50% of the documents fall: its bottom edge is the first quartile (Q1) and its top edge is the third quartile (Q3). The height of the box is therefore the interquartile range (IQR), and the line inside the box is the median.

While the “plain” domain presents a highly compact box with no outliers - highlighting a consistent, predictable uniformity around its median of 16.14 words - the “notice” domain is positioned substantially higher on the scale. Crucially, the “notice” boxplot reveals two clear outliers (indicated by the distinct circular markers): one document with an unusually low average (around 14 words, dipping into the “plain” language range) and another with an exceptionally high average (over 33 words).

This confirms the earlier suspicion: these outlier documents are the primary drivers of the inflated standard deviation noted earlier, even while the core 50% of the “notice” data (the box itself) remains relatively concentrated. Moreover, the lack of vertical overlap between the two interquartile boxes shows that, within this sample, the central sentence lengths of the two domains occupy clearly different ranges.



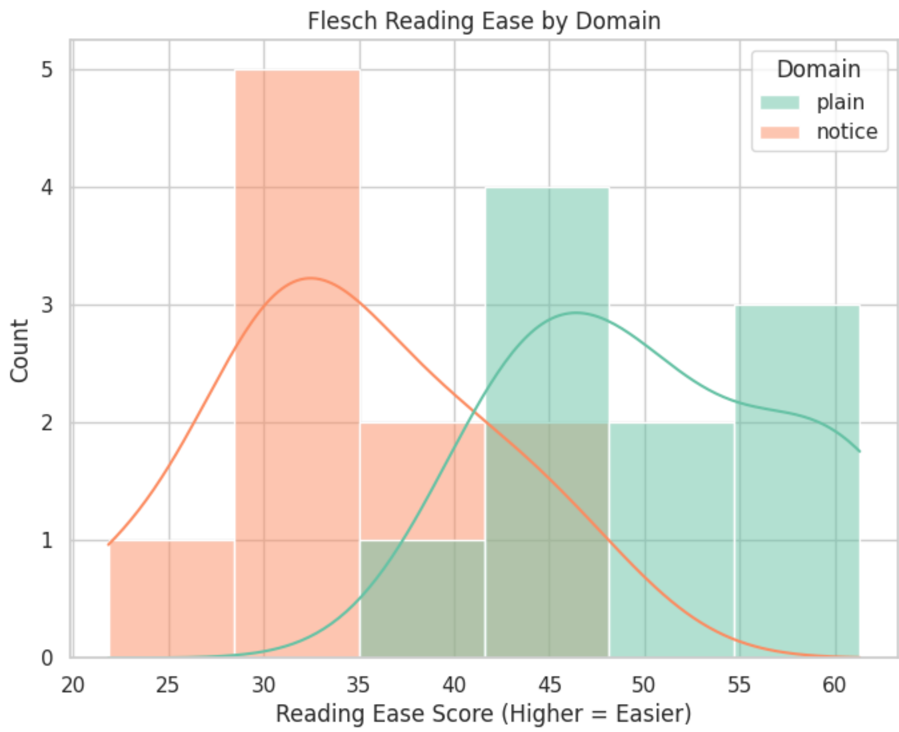

### 2.2. Flesch Reading Ease

The Flesch metric typically ranges from 0 to 100, where higher scores indicate highly accessible text and lower scores reflect dense, structurally complex language. Here, it revealed a substantial divergence in how the two domains convey information. The “plain” domain shows a markedly higher mean score (50.61) compared to the “notice” domain (34.52), indicating that “plain” texts are easier to read. While both domains share an almost identical standard deviation (approximately 7.3), the interquartile range (IQR) for “plain” texts is considerably wider (12.80) than that of the “notice” texts (8.22), pointing to a greater diversity of text styles within the “plain” language category.

The accompanying histogram provides a visual representation of these reading scores. The “notice” domain forms a distinct, slightly skewed distribution shifted toward the lower end of the scale. Most of these texts score between roughly 28 and 35, highlighting a dense, complex writing style typical of official, formal documentation. Its peak is relatively sharp, confirming the tighter IQR.

Conversely, the “plain” language dataset occupies the higher end of the scoring spectrum, reflecting its improved readability. Unlike the sharp peak seen in its sentence length distribution, the Flesch Reading Ease curve for the “plain” domain is notably wider and irregular, spanning from a score of 35 up to 60. This visualizes the wider IQR and suggests that while these texts generally maintain a baseline of accessibility, the specific degree of ease varies more across the “plain” documents than it does within the “notice” documents.

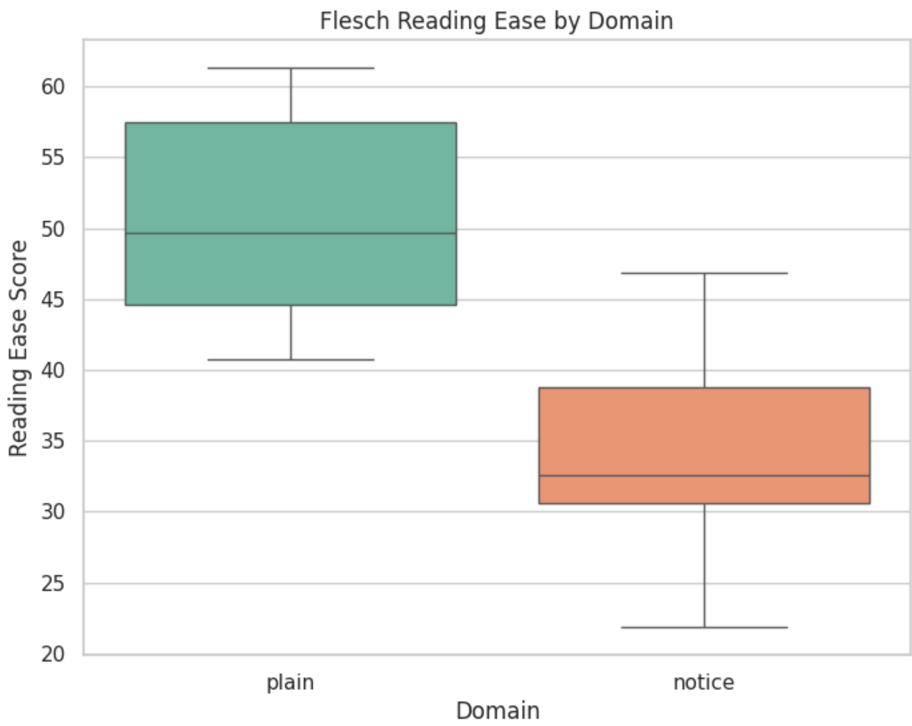
Unlike average sentence length, where the two groups separated almost completely, the readability distributions overlap in the middle of the scale (roughly 35-48), where some “notice” documents reach into the plain texts' readability range. The separation between the domains is therefore real but less clean for readability than for sentence length.




The accompanying boxplot reinforces these observations and clarifies the specific nature of the overlap. Unlike the sentence length's boxplot, neither domain displays any extreme outliers.

The "plain" domain is represented by a noticeably taller box, visually confirming its wider IQR (12.80) and illustrating a broader spread of readability scores around its median of around 50. In contrast, the "notice" domain features a much more compressed box positioned lower on the scale. Centered around a median of approximately 33, it indicates less variation.

Crucially, while the central coloured boxes do not overlap vertically, their whiskers do. The highest "“notice”" scores (around 47) overlap with the lowest "plain" scores (around 40). This confirms that while most texts remain distinct, the easiest official notices occasionally overlap with the least accessible "plain" texts.




3. **Inferential Statistics and Confidence Intervals**

To extend the analysis beyond this specific sample of 20 documents and estimate the true parameters of the broader population, inferential statistics were applied - 95% confidence intervals (CIs) were calculated for the average sentence length of both text categories.

Because of the relatively small sample size (10 documents per domain,), the intervals were constructed using a t-distribution with 9 degrees of freedom ($n - 1$, where $n = 10$ per group). These interval boundaries were built using the sample mean and the standard error (se), which acts as a safety margin to account for the natural unpredictability of sampling data. The intervals provide a range within which we can be 95% confident that the true population mean for average sentence length resides.

The calculated 95% confidence intervals are as follows:

- "Plain" Texts: 14.78 to 17.29 words per sentence
- "Notice" Texts: 21.47 to 28.55 words per sentence

The most critical observation from these results is the complete absence of overlap between the two intervals. The maximum expected average for the “plain” population (17.29 words) falls substantially below the minimum expected average for the “notice” population (21.47 words).

This clear separation between the intervals provides a strong evidence that the difference observed in the descriptive statistics reflects a real difference between the populations, rather than an artifact of random sampling variation. Because the two 95% confidence intervals do not overlap, it is highly unlikely that the two populations share the same mean sentence length. We can therefore infer with a high degree of confidence that corporate privacy notices, at a population level, differ from plain-language documentation in sentence length, relying consistently on longer sentence structures.

## 4. Conclusion

The empirical analysis demonstrates a clear linguistic distinction between corporate privacy notices and the plain-language benchmark. Returning to the initial question of whether privacy notices meet the accessible communication standards outlined by GDPR Article 12, the statistical evidence suggests they do not.

The comparative analysis yields two main findings:

- Syntactic Complexity: Privacy notices consistently rely on longer sentence structures. The non-overlapping 95% confidence intervals provide strong evidence that this difference holds at the population level, rather than being an artifact of the chosen sample.
- Textual Accessibility: While the readability distributions overlap slightly at the margins (the most accessible notices can occasionally match the most complex “plain” texts) the bulk of each distribution remains distinctly separate. Privacy

notices consistently score lower on the Flesch Reading Ease scale, indicating a denser, more formal register.

These conclusions should, however, be read considering the study's main limitation: the analysis’ small sample. The results are therefore indicative rather than definitive. A larger corpus and a formal hypothesis test is necessary to draw definitive conclusions regarding this gap.

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